

How This Project Works

The Complete Tour of the Pokémon-Playing AI Agent

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Abstract

This document explains everything: what the program does, how the AI “sees” and “plays” the game, what each piece of code is responsible for, and what actually happens step by step when you run it. No prior knowledge of the codebase is assumed.

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1 The Big Picture

This project makes an AI language model (LLM) play **Pokémon Red** by itself.

Think of it like this: imagine you are playing Pokémon over a video call with a friend. Your friend cannot touch the Game Boy — they can only:

1. **Look** at a photo of the screen you send them,
2. **Read** a little cheat-sheet you write for them (“you’re in Pallet Town, you have \$3000, no Pokémon yet, there’s a dialog box open…”),
3. **Tell you** which buttons to press (“press A twice, then walk up”).

You press the buttons for them, take a new photo, write a new cheat-sheet, and send it back. Repeat forever.

In this project:

- The “friend” is **Gemini Flash Lite** (an LLM, called through OpenRouter).
- “You” (the one holding the Game Boy) is the **Python program**.
- The Game Boy itself is **PyBoy**, a Game Boy emulator that runs the actual Pokémon Red ROM file on your computer.

One full exchange — *model looks* → *model decides* → *buttons get pressed* → *new screenshot taken* — is called a **step**. You choose how many steps to run:

```
uv run main.py --rom "Pokemon Red.gb" --steps 20
```

2 The Cast of Characters (the Modules)

```
pokemon/
|- main.py                <- entry point: wires everything, starts the loop
|- prompts/              <- the "instructions" given to the AI (Jinja)
|   |- system/game_player.j2  <- "You are playing Pokemon Red..."
|   |- human/summarize_request.j2 <- "Please summarize your progress..."
|   `-- human/summary_handoff.j2 <- "Here is your summary, continue playing..."
`- src/pokemon_agent/
    |- core/
    |   |- settings.py        <- all configuration (model name, temperature...)
    |   `-- llm.py           <- builds the connection to the AI model
    |- agent/
    |   |- graph.py          <- THE BRAIN WIRING: the loop logic (LangGraph)
    |   |- state.py          <- the shared "memory" of the loop
    |   |- tools/press_buttons.py <- the AI's "hands" (button presses)
    |   |- tools/navigate_to.py <- optional auto-walking tool (off by default)
    |   |- vision_tool_node.py <- makes screenshots visible to the AI
    |   `-- prompt_manager.py <- loads & renders the prompt templates
    `-- emulator/
        |- emulator.py       <- wraps PyBoy: buttons, screenshots, maps
        |- memory_reader.py  <- reads the game's RAM and decodes it
        `-- screenshot.py    <- converts screen images to base64 text
```

A quick one-liner for each:

Piece	In plain words
<code>main.py</code>	“Start the Game Boy, build the brain, run N steps.”
<code>settings.py</code>	A single place for every knob (which model, how hot, how many steps...). Reads your <code>.env</code> file.
<code>llm.py</code>	Creates the phone line to Gemini via OpenRouter, with retry logic for when the line is bad.
<code>graph.py</code>	The flowchart the program follows: think $\rightarrow$ act $\rightarrow$ observe $\rightarrow$ repeat $\rightarrow$ summarize when memory is full.
<code>state.py</code>	The clipboard everyone shares: message history, step counter, the emulator itself.
<code>tools/</code>	The two actions the AI is allowed to take.
<code>vision_tool_node.py</code>	The trick that lets the AI literally <i>see</i> screenshots inside tool results.
<code>emulator.py</code>	Knows how to press buttons, take screenshots, and build the collision map.
<code>memory_reader.py</code>	Knows <i>where in the Game Boy’s RAM</i> the game stores money, party, dialog... and decodes it.

3 How the AI “Sees” the Game

The AI never touches pixels or RAM directly. After every action, the program builds an **observation** with three parts (see `agent/tools/_observation.py`):

3.1 The Screenshot

PyBoy exposes the current screen as an image. We:

1. Grab it (`emulator.get_screenshot()` $\rightarrow$ a 160×144 pixel image),
2. Double its size (320×288 — small images are hard for vision models),
3. Encode it as a **base64 data URL** — literally the image converted into a long text string like `data:image/png;base64,iVBORw0KGg...`. LLM APIs only accept text/JSON, so images always travel as text.

3.2 The Memory Cheat-Sheet

The Game Boy stores everything about the game in its RAM, and for a famous game like Pokémon Red, fans have documented **exactly which RAM address holds what**. `memory_reader.py` is a big collection of those addresses plus decoders. Examples:

RAM address	Contains	Quirk
0xD349	Your money	Stored in “Binary Coded Decimal” — each half-byte is one decimal digit
0xD35E	Current location ID	A number 0–255 mapped to names (“PALLET TOWN”, “VIRIDIAN CITY”...)
0xD163	Party size	How many Pokémon you carry
0xD16B+offsets	Pokémon #1 data	Species, HP, level, moves, status...
0xD356	Badges	One bit per badge, 8 badges in one byte
0xC3A0–0xC507	On-screen text	Pokémon uses its own character encoding (0x80 = ‘A’, 0xA0 = ‘a’, ...) which we translate to normal text

The result is a human-readable block the AI receives as text:

Player: RED

Rival: BLUE
Money: \$3000
Location: PALLET TOWN
Coordinates: (5, 6)
Valid Moves: up, down, left
Badges: BOULDER
Inventory:
 POTION x2
Dialog: None

Pokemon Party:
CHARMANDER (CHARMANDER):
Level 8 - HP: 21/25
Types: FIRE
- SCRATCH (PP: 35)
- GROWL (PP: 40)

This matters because vision models are decent but not perfect at reading tiny Game Boy text — memory reading gives the AI *ground truth*.

3.3 The Collision Map (Overworld Only)

When walking around the world, the program also builds an ASCII map of what is walkable:

```
+-----+
|■■■■■■■■|
|■■■■■■■■|
|■■■■■■■■|
|■■■■■■■■|
|■■■■■■■■|
|■■■■■■■■|
|■■■■■■■■|
+-----+
```

- ■ = wall/obstacle (cannot walk there)
- . = walkable path
- S = a sprite (an NPC or object)
- ↓ = **you**, always at the center; the arrow shows which way you are facing

It is built from PyBoy’s “game wrapper”, which exposes the tile grid and collision data of the visible screen (18×20 tiles, which we shrink to 9×10 — one cell per “walkable square”, since the player occupies 2×2 tiles). The facing direction is detected by finding the player’s sprite pattern in the tile grid.

Before the refactor this map was only printed to your terminal. **Now it is sent to the AI too**, so it can plan movement without guessing from pixels.

4 How the AI “Acts”: Tools

LLMs can do **tool calling** (a.k.a. function calling): you describe functions to the model in JSON, and instead of just answering with text, the model can reply “call `press_buttons` with `{"buttons": ["a"]}`”. The program then actually runs that function and sends the result back.

4.1 Tool 1: `press_buttons` (the Main One)

What the model sends:

```
{
  "say": "Time to pick our starter!",
  "buttons": ["up", "up", "a"]
}
```

- **say**: 1–2 sentences spoken aloud to the audience via TTS
- **buttons**: a sequence from the 8 Game Boy buttons (a, b, start, select, up, down, left, right)

What actually happens (in `emulator.py`): for each button, PyBoy holds the button for 10 frames, releases it, then runs the game until it *settles*. Settle detection polls every 8 frames and declares the action finished when (a) the game engine reports idle (walk counter zero, no scripted inputs — documented RAM flags from the pokered disassembly) AND (b) the game’s own screen-composition buffer (`wTileMap` at `0xC3A0`, with the blinking masked) stays byte-identical for 4 consecutive samples (10 samples when the textbox area is blank, the signature of a scene-transition pause). A 400-frame ceiling covers states that never go idle (cutscene walks, battle intros). Measured against the old fixed 120-frame wait: walking settles in ~ 40 frames ($3\times$ faster), menus faster still, and dialogs wait exactly as long as the text takes to print — the fixed wait used to capture half-printed dialog $\sim 16\%$ of the time; settle detection captures complete text.

What the model gets back:

```
{
  "result": "Pressed buttons: up, up, a",
  "memory_info": "Player: RED\nMoney: $3000\nLocation: ...",
  "collision_map": "+-----+\n|■■■■■■■■■■|.....|..."
}
```

plus the screenshot as an actual image (see Section 6 — this is the `VisionToolNode` trick).

4.2 Tool 2: `navigate_to` (Optional, Off by Default)

Instead of pressing one arrow at a time, the model can say “walk me to grid cell (row 2, col 7)”. The program then runs the **A* pathfinding algorithm** over the 9×10 collision grid:

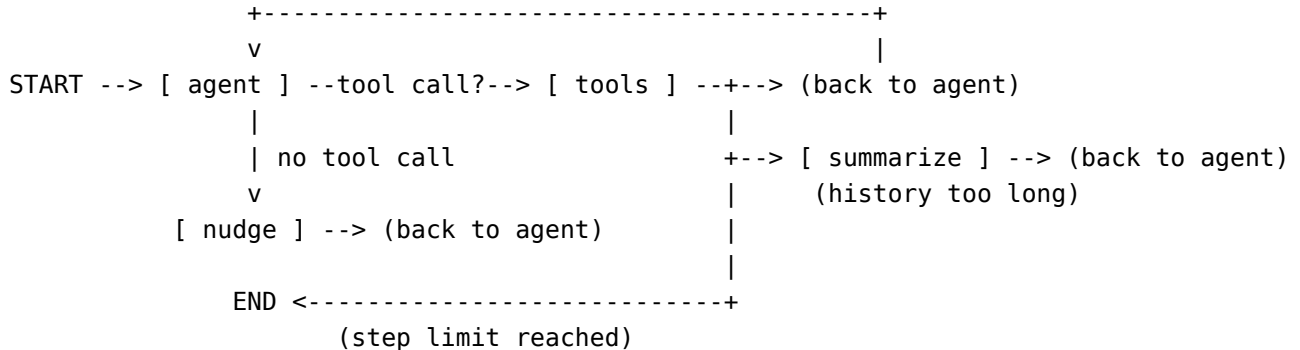
1. Start at the player position (always the center, (4, 4)).
2. Explore neighboring walkable cells, always trying the most promising path first (“promising” = steps taken so far + straight-line distance remaining).
3. Avoid walls, avoid sprites, and respect Pokémon’s weird “tile-pair collision” rules (special cases like ledges in caves/forests where you cannot step from tile X to tile Y even though both are walkable).
4. If the target is unreachable, walk to the closest reachable cell instead.

It returns the list of moves (“up”, “up”, “right”, ...) and presses them one by one. Enable it with `USE_NAVIGATOR=true` in your `.env`.

Why off by default? It only sees the current screen (a 9×10 window), so it cannot plan across screens anyway — and forcing the model to move step by step keeps it engaged with what is happening.

5 The Brain Wiring: the LangGraph Flow

The loop is built with **LangGraph**, a library where you define your agent as a flowchart (“graph”) of **nodes** (boxes that do work) connected by **edges** (arrows, possibly conditional). The flowchart here:



5.1 The Shared Clipboard: `GameAgentState`

Every node reads and writes one shared state object:

```
{
  "messages": [...],      # the full conversation with the AI so far
  "emulator": <Emulator>, # the live Game Boy (tools grab it from here)
  "step_count": 3,         # how many actions taken so far
  "max_steps": 10,         # stop after this many
  "max_history_tokens": 30000, # summarize when a prompt reaches this size
}
```

messages is special: it uses LangGraph’s **add_messages** rule, meaning nodes do not overwrite the list — they *append* to it (or delete from it with special “remove” markers, used by summarization).

5.2 Node by Node

agent — sends the whole conversation to the model and gets back its decision. By default (`FORCE_TOOL_USE=false`) the tools are bound with `tool_choice="auto"`, so the model may narrate its reasoning before acting; a reply without any tool call is handled by the **nudge** node. Setting `FORCE_TOOL_USE=true` switches to `tool_choice="any"`, which **forces** a tool call every turn — never stalls, but (by API design) suppresses all visible reasoning text. If the reply contains a tool call, the step counter goes up by one.

tools — actually executes the tool call (presses the buttons), captures the observation (screenshot + memory + collision map), and appends it to the conversation as a “tool result” message.

summarize — see Section 7.

nudge — handles replies with no tool call (a tool-less reply means nothing happened in the game). This node appends a human message — *“You did not call a tool in your last reply. Nothing happened in the game...”* (from `prompts/human/nudge_no_tool.j2`) — and sends the conversation back to the agent for another try. The message matters for two reasons: it keeps the conversation alternating user/assistant (providers dislike two assistant messages in a row), and it gives the model an explicit signal to break out of a “talking instead of acting” pattern. There is no retry cap — the run only ends at the step limit; LangGraph’s `recursion_limit` (set in `main.py`) is the final cost backstop if a model never acts.

With the default `FORCE_TOOL_USE=false` (`tool_choice="auto"`), the nudge is the active stall-prevention. With `FORCE_TOOL_USE=true`, tool calls are forced at the API level and this node is nearly dead code (it would only fire on provider quirks or truncated replies).

5.3 The Routing Decisions (the Arrows)

After **agent**: did the reply contain a tool call? Yes → go to **tools**. No → **nudge**, then back to the agent. The agent side never ends the run.

After **tools**, three checks in order:

1. `step_count >= max_steps?` → **END** (we are done; you asked for N steps).
2. Did the last turn's prompt reach `max_history_tokens`? → **summarize** (conversation getting too fat). The size comes from the provider's reported token usage — the exact measured size of the conversation, images included — not from counting messages or running a local tokenizer. (If a provider returns no usage data, a 30-message count is the fallback trigger.)
3. Otherwise → back to **agent** for the next step.

6 The Conversation Itself (What the AI Literally Receives)

LLMs are stateless — they remember *nothing* between calls. Every single step, we send the **entire conversation from the beginning**. The conversation looks like:

```
[SYSTEM] You are playing Pokemon Red... (from prompts/system/game_player.j2)
[USER]    You may now begin playing.
[AI]      I'll start by pressing A to begin. -> press_buttons(["a"])
[TOOL]    {"result": "Pressed buttons: a", "memory_info": ...} + <screenshot>
[AI]      A dialog is open, I'll continue. -> press_buttons(["a"])
[TOOL]    {"result": ...} + <screenshot>
[AI]      ...                                -> press_buttons(["a", "a"])
...
```

Reality check: how much useful text the [AI] line contains depends on the model and on `FORCE_TOOL_USE`. Observed in real runs:

- `FORCE_TOOL_USE=true` (`tool_choice="any"`): *no* text at all, from any model — forced tool use suppresses text by API design. The conversation is [AI] "" -> `press_buttons(...)`.
- `auto` + Claude Haiku 4.5: proper narration every turn (“I see the main menu with NEW GAME and OPTION. I’ll select NEW GAME...”).
- `auto` + Gemini Flash Lite: acts reliably but emits only stray token fragments (“elf”, “thought”) instead of real explanations — a quirk of small Gemini models when mixing text with function calls. Harmless, just not informative.

A mandatory “reasoning” parameter in the tool schema would force explanations out of any model, at the cost of a few tokens per turn.

6.1 The VisionToolNode Trick

There is a subtlety here. In the OpenAI-style API format, **tool result messages normally carry only text** — no images. The naive workaround (which this project used before the refactor) was to send the screenshot as a separate follow-up user message, which clutters the conversation.

VisionToolNode does it properly. When a tool returns a dictionary, it checks for image-ish keys (like "screenshot") containing a `data:image/...` string, and splits the tool message into **content blocks**:

```
[
  {"type": "text",      "text": "{\"result\": \"Pressed buttons: a\", ...}\""},
  {"type": "image_url", "image_url": {"url": "data:image/png;base64,iVBOR..."}}
]
```

LangChain supports image blocks inside tool messages, and OpenRouter translates this correctly for Gemini. So the screenshot rides *inside* the tool result, exactly where it belongs.

7 Summarization: the Memory Diet

The problem: every step adds ~2 messages, and each tool message contains a full screenshot (~50–100 KB of base64 text). After 15 steps you are sending megabytes of history on *every* call. Cost grows, speed drops, and eventually you would blow past the model’s context limit.

The fix: when a turn’s prompt reaches `max_history_tokens` (default 30,000, measured from the provider’s reported usage), the `summarize` node:

1. Sends the whole conversation to the model with one extra request: *“Summarize your progress: key events, decisions, current objectives, location, team status, plans.”* (from `prompts/human/summarize_request.j2`)
2. Wipes the **entire** message history (a special `REMOVE_ALL_MESSAGES` marker).
3. Rebuilds a tiny fresh conversation:
 - the same system prompt,
 - one user message: *“CONVERSATION HISTORY SUMMARY (representing 30 previous messages): ...”* + a **fresh screenshot** so the model is not blind + *“You may now continue playing.”*

The model loses the play-by-play details but keeps the plot. It is like a “previously on...” TV recap. Then the loop continues normally.

8 Two Full Scenarios, Frame by Frame

8.1 Scenario A: a Normal Step (Talking to Professor Oak)

1. **agent node** → the entire conversation (system prompt + history) is POSTed to OpenRouter. Gemini sees the last screenshot showing a dialog box: *“OAK: Your very own POKÉMON legend is about to unfold!”*
2. Gemini replies: text reasoning *“There’s an active dialog, I should advance it”* + tool call `press_buttons(buttons=["a"], wait=true)`.
3. **Router:** tool call exists → go to **tools**.
4. **tools node** → grabs the emulator from state; PyBoy holds A for 10 frames, releases, then runs until the game settles (the dialog finishes printing). Then it captures: new screenshot (base64), memory dump (`Dialog: "So! You want the fire POKÉMON, CHARMANDER?"`), no collision map (we are in a dialog, not the overworld).
5. The observation becomes a multimodal tool message appended to the conversation. Step counter: 7 → 8.
6. **Router:** $8 < \text{max_steps}$, $18 \text{ messages} < 30$ → back to **agent**. Repeat.

8.2 Scenario B: Hitting the History Limit Mid-Game

1. Step 14 finishes; the last prompt weighed in at 31,000 tokens. Router after **tools**: step limit not reached, but $31,000 \geq \text{max_history_tokens}$ $\rightarrow$ go to **summarize**.
2. The summarize node asks the model for a recap. Gemini returns: *“1. Key events: started the game, chose Charmander... 4. Currently on Route 1 heading to Viridian City...”*
3. History is wiped. New history = system prompt + recap + fresh screenshot + “continue playing”. That is 2 messages instead of 30.
4. Back to **agent**: Gemini reads the recap, looks at the screenshot, and carries on walking north — without remembering each individual button press, but knowing exactly where it is and why.

9 The Connection to the AI: `llm.py`

We talk to **OpenRouter** — a broker that exposes hundreds of models behind one OpenAI-compatible API. Which model is just a setting (`GAME_MODEL=google/gemini-3.1-flash-lite`), so you can swap to Claude or GPT by editing `.env`, no code changes.

`get_llm()` builds a **RobustChatOpenAI** — a hardened version of LangChain’s standard OpenAI client (ported from the `tolatex` backend) that handles real-world OpenRouter problems:

- **Garbage in a 200 response**: sometimes the HTTP call “succeeds” but the body is truncated JSON, or contains an embedded error (`{"message": "Provider returned error", "code": 500}`). It detects these and retries with growing pauses (3s, 9s, 27s).
- **Infinite generation loops**: if the model gets stuck repeating itself until it hits the token ceiling (`finish_reason: "length"`), it can detect this and retry (off by default).
- **Reasoning preservation**: “thinking” models return their reasoning in nonstandard fields that normally get dropped; this client saves them and re-sends them on later turns so multi-turn thinking stays coherent. With `THINKING=none` (the default here) this is dormant, but flip `THINKING=low/medium/high` in `.env` and it activates.
- **Provider sorting**: requests OpenRouter’s lowest-latency provider for the model.

10 Configuration: `settings.py` and `.env`

All knobs live in one pydantic-settings class. Anything can be set in `.env` (or as a real environment variable, case-insensitive):

Setting	Default	Meaning
OPENROUTER_API_KEY	—	your key (required)
GAME_MODEL	google/gemini-3.1-flash-lite	the model that plays
SUMMARIZER_MODEL	"" (= game model)	the model that writes recaps
TEMPERATURE	1.0	randomness of decisions
MAX_TOKENS	4000	max length of each model reply
THINKING	none	reasoning effort: none/minimal/low/medium/high
MAX_STEPS	10	default number of steps (CLI --steps overrides)
MAX_HISTORY_TOKENS	30000	prompt size that trig- gers a summary (CLI --max-history-tokens over- rides)
USE_NAVIGATOR	false	expose the A* navigate_to tool
FORCE_TOOL_USE	false	true forces a tool call every turn but suppresses reasoning text
ROM_PATH	Pokemon Red.gb	default ROM file (CLI --rom overrides)
MAX_REQUESTS_PER_SECOND	0 (off)	global API rate limit

`get_settings()` is cached — the `.env` is read once per run. The prompts are equally swappable: they are Jinja2 template files in `prompts/`, so you can tweak the AI’s personality or goals without touching Python.

11 Money: What a Run Costs

Each step sends the full history, so cost grows until a summary resets it. With Gemini Flash Lite, observed costs were roughly **\$0.0001–0.0025 per step** (rising as history grows), so a 10-step run is about a penny. The two real cost levers:

- **MAX_HISTORY_TOKENS** — lower = more frequent summaries = cheaper but more forgetful.
- The model — flash-lite is among the cheapest vision models; switching to a frontier model multiplies cost by $\sim 100\times$.

12 Honest Limitations (So You Are Not Surprised)

- **One action per turn, no multi-reply deliberation** — the loop expects each reply to act (tool-less replies just get nudged); the model cannot “stop and think” across several replies before committing.
- **It is myopic**: it sees one screen at a time and a 9×10 collision window. Long-range navigation (cross-town trips) relies on its world knowledge of Pokémon Red, which is decent but imperfect.
- **Summaries lose detail**: if something important was not mentioned in the recap, it is gone (e.g., “I already searched that house”).
- **Flash-lite is cheap, not brilliant**: it tends to skip the “explain your reasoning” part and just mash A through dialogs. It works, but do not expect speedrun strategy. Swap in

a stronger model for smarter play.

- **No save/checkpoint logic in the agent** — the emulator supports loading a saved state (`--load-state file`), but the agent does not auto-save progress.